

Yannis Tsividis

professor

基本信息

姓名: Yannis Tsividis

所属院校: 哥伦比亚大学

地区: United States

年龄范围: 50岁以上

企业经历: 有

荣誉称号

★ Member of the National Academy of Engineering

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

电子信息工程

电子与计算机工程

集成电路设计与集成系统

微电子科学与工程

研究领域

信号与信息处理

数字系统设计与FPGA

低功耗电路与能量管理

计算机体系结构

低功耗设计与能耗优化

个人简介

Yannis Tsividis, Edwin Howard Armstrong Professor of Electrical Engineering at Columbia University, is renowned for his pioneering work in analog and digital circuit integration, analog computing, and hybrid computing on silicon chips. His contributions include the development of the first fully-integrated MOS operational amplifier and the widely-used Switcap software. Tsividis has

authored numerous patents and a seminal textbook on MOS transistors. His accolades include the IEEE Gustav Robert Kirchhoff Award and the IEEE Circuits and Systems Education Award. A member of the US National Academy of Engineering, Tsividis has significantly impacted the field of electrical engineering through his research and education.

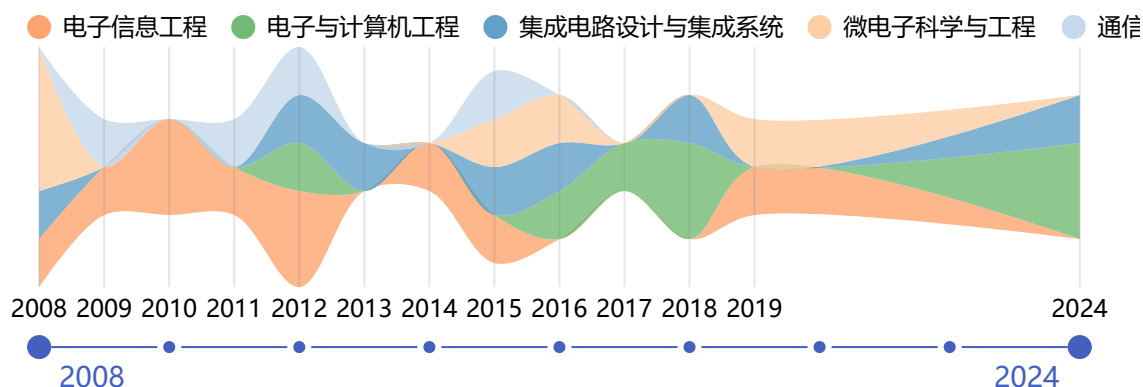
工作经历

Columbia University - Edwin Howard Armstrong Professor of Electrical Engineering	2014 - 至今
Columbia University - Charles Batchelor Professor of Electrical Engineering	1998 - 2014
University of Paris Marie Curie (now part of Sorbonne universities) - Visiting Professor	- 2008
National Technical University of Athens - Professor	1990 - 1994
Columbia University - Assistant Professor, Associate Professor, and Professor	1976 - 1998
National Technical University of Athens, Greece - Visiting Professor	- 1983
Massachusetts Institute of Technology - Visiting Assistant Professor	- 1980
AT&T Bell Laboratories, Murray Hill - Resident Visitor	1977 - 1987

教育经历

University of Minnesota, Minneapolis - B.E.E.	- 1972
University of California, Berkeley - M.S.	- 1973
University of California, Berkeley - Ph.D.	- 1976

近年发表的论文



SPADES: A 0.54-GFLOPS/W Sparse Matrix Vector Multiplication Accelerator Featuring On-the-Fly GZIP Decompression for 3.36x Reduction in Off-Chip Data Movement

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Huang, Paul Xuanyuanliang; Tsvidis, Yannis; Seok, Mingoo

被引用次数: 0

2024

AACIM: A 2785-TOPS/W, 161-TOP/mm², <1.17%-RMSE, Analog-In Analog-Out Computing-In-Memory Macro in 28nm

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Wan, Rentao; Lin, Chuan-Tung; Zhang, Bo; Tsvidis, Yannis; Seok, Mingoo

被引用次数: 0

2024

INTIACC: A Programmable Floating-Point Accelerator for Partial Differential Equations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Huang, Paul Xuanyuanliang; Tsvidis, Yannis; Seok, Mingoo

被引用次数: 2

2024

Signal Encoding and Processing in Continuous Time Using a Cascade of Digital Delays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Patil, Sharvil; Rao, Suhas Gundu; Chen, Yu; Tsvidis, Yannis

被引用次数: 6

2019

NOT YOUR FATHER'S ANALOG COMPUTER

IEEE SPECTRUM

Tsvidis, Yannis

被引用次数: 31

2018

Energy-Efficient Hybrid Analog/Digital Approximate Computation in Continuous Time (vol 51, pg 1514, 2016)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Ning; Huang, Yipeng; Mai, Tao; Patil, Sarvil; Cao, Chi; Seok, Mingoo; Sethumadhavan, Simha; Tsvidis, Yannis

被引用次数: 1

2018

A Continuous-Time Digital IIR Filter With Signal-Derived Timing and Fully Agile Power Consumption

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Yu; Zhang, Xiaoyang; Lian, Yong; Manohar, Rajit; Tsividis, Yannis

被引用次数: 13

2018

ANALOG COMPUTING IN A MODERN CONTEXT: A LINEAR ALGEBRA ACCELERATOR CASE STUDY

IEEE MICRO

Huang, Yipeng; Guo, Ning; Seok, Mingoo; Tsividis, Yannis; Sethumadhavan, Simha

被引用次数: 29

2017

Energy-Efficient Hybrid Analog/Digital Approximate Computation in Continuous Time

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Ning; Huang, Yipeng; Mai, Tao; Patil, Sharvil; Cao, Chi; Seok, Mingoo; Sethumadhavan, Simha; Tsividis, Yannis

被引用次数: 67

2016

A 3-10 fJ/conv-step Error-Shaping Alias-Free Continuous-Time ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Patil, Sharvil; Ratiu, Alin; Morche, Dominique; Tsividis, Yannis

被引用次数: 20

2016

Improving the Energy Efficiency of Pipelined Delay Lines Through Adaptive Granularity

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Vezyrtzis, Christos; Tsividis, Yannis; Nowick, Steven M.

被引用次数: 0

2015

Derivative Level-Crossing Sampling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Martinez-Nuevo, Pablo; Patil, Sharvil; Tsividis, Yannis

被引用次数: 25

2015

A Flexible, Event-Driven Digital Filter With Frequency Response Independent of Input Sample Rate

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Vezyrtzis, Christos; Jiang, Weiwei; Nowick, Steven M.; Tsividis, Yannis

被引用次数: 24

2014

An Event-driven Clockless Level-Crossing ADC With Signal-Dependent Adaptive Resolution

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Weltin-Wu, Colin; Tsividis, Yannis

被引用次数: 105

2013

A detailed look at electrical equivalents of uniform electrochemical diffusion using nonuniform resistance-capacitance ladders

JOURNAL OF ELECTROANALYTICAL CHEMISTRY

Tsividis, Yannis; Milios, John

被引用次数: 29

2013

Event-Driven GHz-Range Continuous-Time Digital Signal Processor With Activity-Dependent Power Dissipation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kurchuk, Mariya; Weltin-Wu, Colin; Morche, Dominique; Tsividis, Yannis

被引用次数: 37

2012

A Case for Hybrid Discrete-Continuous Architectures

IEEE COMPUTER ARCHITECTURE LETTERS

Sethumadhavan, Simha; Roberts, Ryan; Tsividis, Yannis

被引用次数: 7

2012

Internally Non-LTI Systems Based on Delays, With Application to Companding Signal Processors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mai, Tao; Tsividis, Yannis

被引用次数: 0

2012

Externally Linear Discrete-Time Systems With Application to Instantaneously Companding Digital Signal Processors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Klein, Aaron; Tsividis, Yannis

被引用次数: 3

2011

Signal-Dependent Variable-Resolution Clockless A/D Conversion With Application to Continuous-Time Digital Signal Processing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kurchuk, Mariya; Tsividis, Yannis

被引用次数: 68

2010

Externally Linear Time Invariant Digital Signal Processors

IEEE TRANSACTIONS ON SIGNAL PROCESSING

Klein, Aaron E.; Tsividis, Yannis

被引用次数: 3

2010

Analysis and simulation of continuous-time digital signal processors

SIGNAL PROCESSING

Schell, Bob; Tsividis, Yannis

被引用次数: 10

2009

A low power tunable delay element suitable for asynchronous delays of burst information

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Schell, Bob; Tsividis, Yannis

被引用次数: 37

2008

A 2.4-GHz ISM-band sliding-IF receiver with a 0.5-V supply

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Stanic, Nebojsa; Balankutty, Ajay; Kinget, Peter R.; Tsividis, Yannis

被引用次数: 43

2008

A Continuous-Time ADC/DSP/DAC System With No Clock and With Activity-Dependent Power Dissipation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Schell, Bob; Tsividis, Yannis

被引用次数: 147

2008

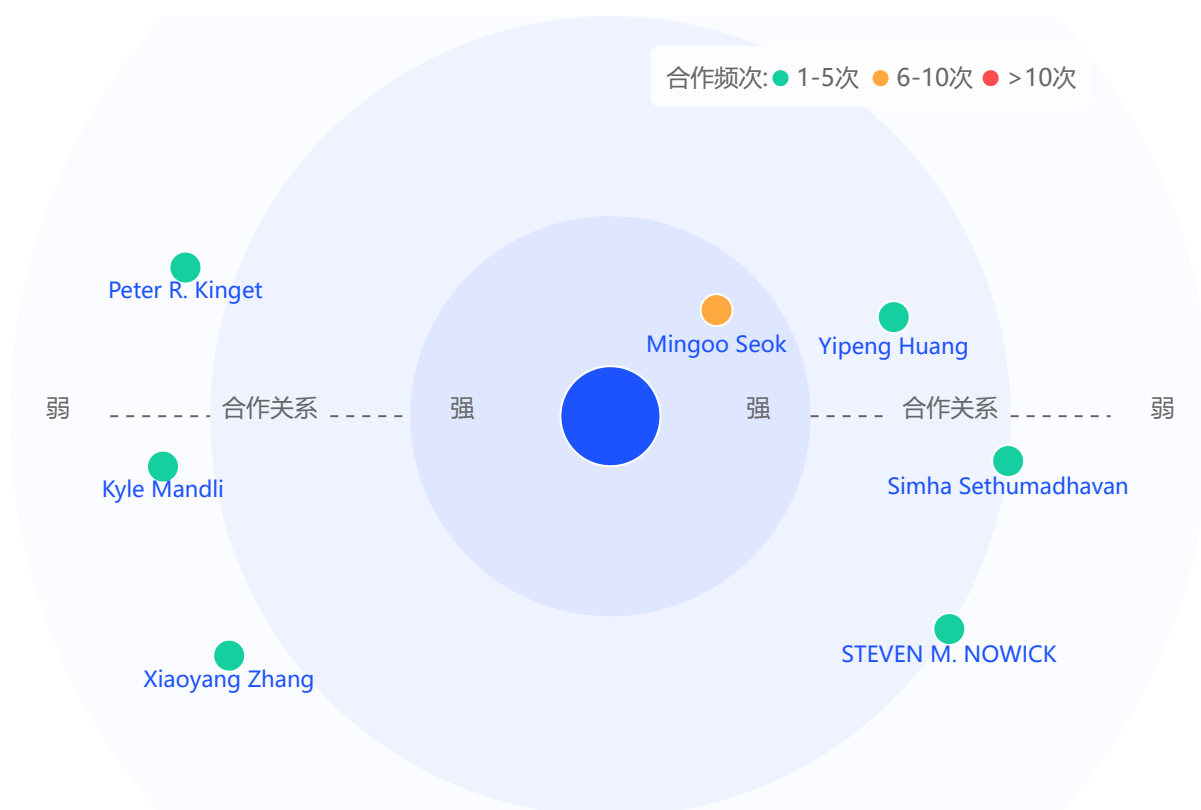
Hybrid Analog-Digital Solution of Nonlinear Partial Differential Equations

MICRO

Yipeng Huang; Ning Guo; Mingoo Seok; Yannis Tsividis; Kyle Mandli; Simha Sethumadhavan

被引用次数: 0

学术合作



Mingoo Seok

哥伦比亚大学

合作次数: 7

Yipeng Huang 罗格斯大学-新布朗斯维克	合作次数: 4
Simha Sethumadhavan 哥伦比亚大学	合作次数: 2
STEVEN M. NOWICK 哥伦比亚大学	合作次数: 2
Xiaoyang Zhang 新加坡国立大学	合作次数: 1

还有 2 个合作者